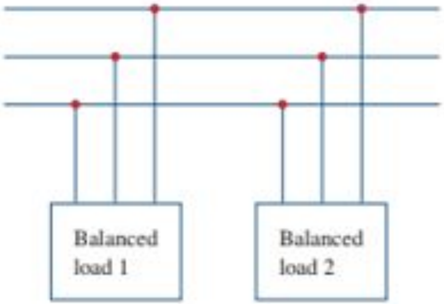
Practice Problem 12.8

Assume that the two balanced loads in Fig. 12.22(a) are supplied by an 840-V rms 60-Hz line. Load 1 is Y-connected with $30 + j40\ \Omega$ per phase, while load 2 is a balanced three-phase motor drawing 48 kW at a power factor of 0.8 lagging. Assuming the *abc* sequence, calculate: (a) the complex power absorbed by the combined load, (b) the kVAR rating of each of the three capacitors Δ-connected in parallel with the load to raise the power factor to unity, and (c) the current drawn from the supply at unity power factor condition.



(a) given for load-1

$V_l = 840\ V$

$Z_\Delta = 30 + j40$

$\therefore \bar{S}_1 = \dfrac{3V_p^2}{(30 - j40)}$ $\quad \Big|\ V_p = \dfrac{840}{\sqrt{3}}$

$= 8.5k + j11.3k$

For load. 2

$$P_{T2} = 48\,kW$$

$$pf_2 = 0.8$$

$$P_{T_2} = 3V_pI_p\cos\theta$$

$$S_{T_2} = 3V_p I_p = \frac{P_{L2}}{0.8}$$

$$= 60\,kVA$$

$$Q_{T_2} = S_{T_2} \sin\theta$$

$$= 60k \times \sin(37^\circ)$$

$$= 36.12\,k$$

$$\theta = \cos^{-1}(0.8)$$

$$= 37^\circ$$

$$\bar{S}_{T_2} = P_{T_2} + jQ_{T_2}$$

$$= 48k + j\,36.12\,k$$

$$\bar{S} = \bar{S}_{T_1} + \bar{S}_{T_2}$$

$$= (56.5k + j\,47.42k)\ VA$$

(b) From (a) we et,

$Q = 47.42k\ VAR$

[To make pf = 1, $Q_2 = 0$]

change in Q,

$Q_C = Q - Q_2$

$= Q - 0$

$= Q$

$\therefore Q_C = Q$

So the KVAR rating of the cap,

$$Q_{cap} = \frac{Q_C}{3}$$

$$= 15.81\ KVAR$$

(C)

$$\therefore S_T = P_T + jQ_T$$

$$= P_T + j0$$

$$= 56.5\,kW + 0$$

$$= 56.5\,kW$$

$=$

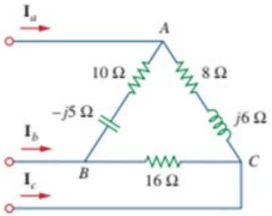
$$\bar{S}_T = \sqrt{3}\, V_L I_L \angle\theta$$

$$= \sqrt{3}\, V_L I_L \angle 0^\circ$$

$$\theta = \cos^{-1}(1) = 0^\circ$$

$$\therefore I_L = \frac{\bar{S}_T}{\sqrt{3}\, V_L} = \frac{56.5k}{\sqrt{3} \times 840} = 38.834A$$

Practice Problem 12.9



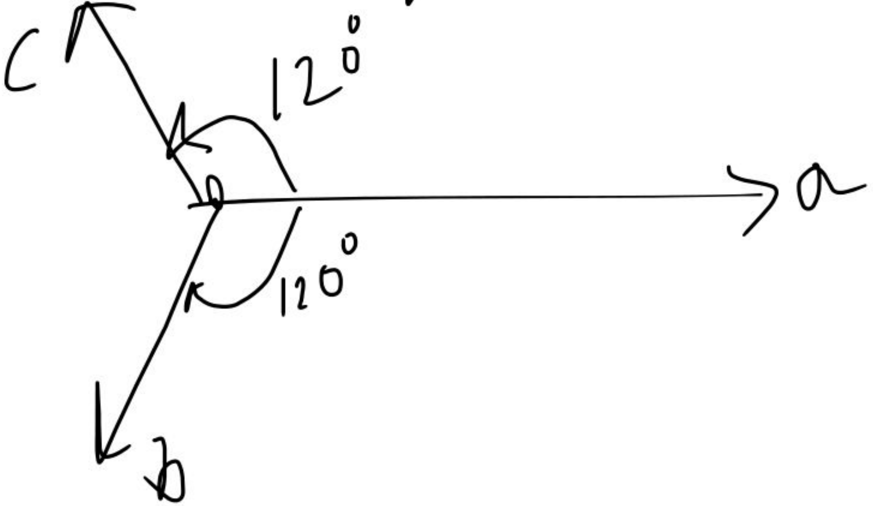
The unbalanced Δ-load of Fig. 12.24 is supplied by balanced line-to-line voltages of 440 V in the positive sequence. Find the line currents. Take $\mathbf{V}_{ab}$ as reference.

$V_{ab} = 440\angle 0^\circ$

$V_{bc} = 440\angle -120^\circ$

$V_{ca} = 440\angle 120^\circ$

Positive sequence



Here, $V_{\phi_{load}} = V_{ab}$

$V_{ca} = 440\angle 180+120$

$$\therefore I_a = \frac{V_{ab}}{10-j5} + \frac{V_{ac}}{8+j6}$$

$$= \frac{440\angle 0^\circ}{10-j5} + \frac{440\angle 30^0}{8+j6}$$

$$= 39.71 \angle -41.06^0$$

$$I_b = \frac{V_{bc}}{16} + \frac{V_{ba}}{10-5j} \quad \Big| \; V_{ba} = 440\angle 180^\circ$$

$$= \frac{440\angle -120^\circ}{16} + \frac{440\angle 180^\circ}{10-5j}$$

$$= -48.95 - j\,41.42 = 64.12\angle -140^\circ$$

$$I_c = \frac{V_{cb}}{16} + \frac{V_{ca}}{8+6j}$$

$$= \frac{440\angle 60^\circ}{16} + \frac{440\angle 120^\circ}{8+6j}$$

$$= 70.13\angle 79.27^\circ$$

$$V_{cb} = -V_{bc}$$

$$= 440\angle(180^\circ - 120^\circ)$$

$$= 440\angle 60^\circ$$

Calculate the line current required for a 30-kW three-phase motor having a power factor of 0.85 lagging if it is connected to a balanced source with a line voltage of 440 V.

Practice Problem 12.7

here.

$P = \sqrt{3}\, V_L I_L \cos\theta$

$\therefore I_L = \dfrac{P}{\sqrt{3} \times V_L \times \cos\theta}$

$= 46.31\,A$

$P = 30\,KW$

$V_L = 440$

$Pf = \cos\theta$

$= 0.85$

Practice Problem 12.6

For the Y-Y circuit in Practice Prob. 12.2, calculate the complex power at the source and at the load.

Practice Problem 12.2

A Y-connected balanced three-phase generator with an impedance of $0.4 + j0.3\ \Omega$ per phase is connected to a Y-connected balanced load with an impedance of $24 + j19\ \Omega$ per phase. The line joining the generator and the load has an impedance of $0.6 + j0.7\ \Omega$ per phase. Assuming a positive sequence for the source voltages and that $\mathbf{V}_{an} = 120\underline{/30^\circ}$ V, find: (a) the line voltages, (b) the line currents.

Sol^n

$$I_p = \frac{120\angle 30^\circ}{(0.4+0.3j)+(0.6+0.7j)+(24+j19)}$$

$$= 3.71 - 0.56j$$

$$|I_p| = 3.75$$

As the connection is Y—Y so the same current I_p will flow through the source phase & load phase.

Complex power at source

$$= -3 \times \bar{V}_p \times \bar{I}_p^*$$

$$= -3 \times (120 \angle 30^\circ) \times (3.71 + 0.56j)$$

$$= -(1055.87 + 842.4j) \text{ VA}$$

Power at the load.

$$S_L = 3 \times |I_p|^2 \times Z_L$$

$$= 3 \times (3.75)^2 \times (24 + j19)$$

$$= (1013.52 + 802.37j)\ VA$$